EE3032 Introduction to VLSI Design—Homework 2

1. **Multi-V_t Design** (3 points)
   Fig. 1(a) shows a combinational logic in which each logic gate has 1 unit delay if low V_t transistors are used to realize the logic gates. If we want to reduce the subthreshold conduction power of the circuit using dual-V_t process (i.e., you have high-V_t and low-V_t transistors) without sacrificing the performance of the circuit, please draw the transistor-level schematic of the combinational circuit with dual-V_t transistors using the symbols shown in Fig. 1(b).

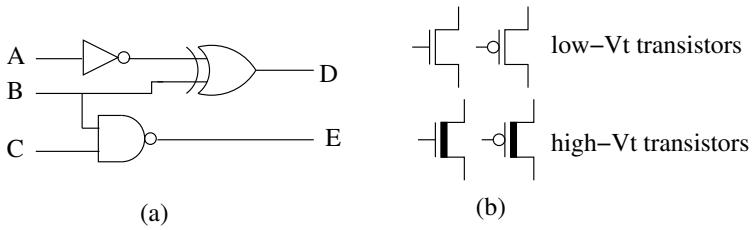

Figure 1: (a) A simple CMOS logic (b) Symbols of transistors

2. **Non-ideal Effect** (3 points)
   Fig. 2 shows a simple circuit with a capacitance. Assume that the threshold voltage and the pn-junction leakage current of the NMOS transistor are V_t and I_L, respectively. Also, the switching threshold voltage of the buffer is $V_{dd}/2$. Complete the following questions.

   (a) If we turn on the NMOS by asserting SL=1 and precharging the capacitance through the PMOS transistor by asserting Pre=0, what voltage level of the capacitance can be precharged to? (1 point)

   (b) If we turn off the NMOS and PMOS, how long the logic 1 (output still can read data 1) can be retained? (2 points)

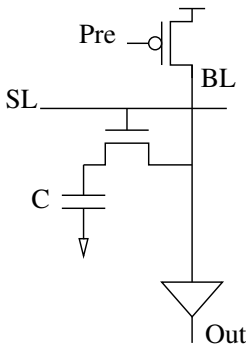

Figure 2: A simple circuit with a capacitance.

3. **Logic Gate** (4 points)
   Fig. 3 shows a CMOS logic gate. If the size of NMOS and PMOS transistors is W/L, the corresponding turned-on resistance is R_n and R_p, respectively, regardless of the transistor is operated at linear or saturation regions. Also, $R_p = 2R_n$, where $\mu_n = 2\mu_p$ is assumed. Complete the following questions.

(a) What is the Boolean function realized by the logic gate? (1 points)

(b) If the smallest voltage of logic 0 is $0.1V_{dd}$, determine the sizes of the transistors such that the logic gate can operate correctly. (3 points)

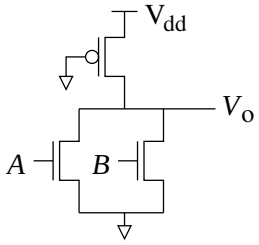

Figure 3: CMOS logic gate